

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12405252
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Chen; Lijun et al.

Method for detecting contents of whey protein and casein, and/or ratio thereof in milk powder

Abstract

The present disclosure relates to a method for detecting the contents of whey protein and casein, and/or the ratio thereof in milk powder by liquid chromatography-mass spectrometry. According to the method of the present disclosure, the characteristic peptide segments of whey protein and casein are selected and used for correction, and thus the accuracy and stability of detection of peptide segments by mass spectrometry using external standards are improved; in addition, the method of the present disclosure has the following advantages: easy to pretreat samples, low cost, a wide range of applicability, enable to perform quantitative detection of whey protein at one time, not affected by protein denaturation in the production process, no need to synthesize isotope internal standard, and avoiding influence of ionization efficiency and matrix in mass spectrometry detection by means of their own multiple characteristic peptide segment ratios.

Inventors: Chen; Lijun (Beijing, CN), Xu; Tao (Beijing, CN), Chen; Jingyao (Beijing, CN), Qiao; Weicang (Beijing, CN), Zhao; Junying (Beijing, CN)

Applicant: Beijing Sanyuan Foods Co., Ltd. (Beijing, CN)

Family ID: 77481046

Assignee: BEIJING SANYUAN FOODS CO., LTD. (Beijing, CN)

Appl. No.: 17/871549

Filed: July 22, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230080697 A1	Mar. 16, 2023

Foreign Application Priority Data

CN	202110904028.8	Aug. 06, 2021
----	----------------	---------------

Related U.S. Application Data

continuation parent-doc WO PCT/CN2021/143234 20211230 PENDING child-doc US 17871549

Publication Classification

Int. Cl.: **G01N30/06** (20060101); **G01N30/30** (20060101); **G01N30/34** (20060101); **G01N30/72** (20060101); **G01N30/88** (20060101); G01N33/04 (20060101)

U.S. Cl.:

CPC **G01N30/06** (20130101); **G01N30/30** (20130101); **G01N30/34** (20130101); **G01N30/72** (20130101); **G01N30/88** (20130101); G01N33/04 (20130101)

Field of Classification Search

CPC: G01N (30/06); G01N (30/30); G01N (30/34); G01N (30/72); G01N (30/88); G01N (33/04); G01N (2030/8831); G01N (30/02); G01N (30/32); G01N (30/8675); G01N (2030/324)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2021/0088515	12/2020	Paullin	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
108152385	12/2017	CN	N/A
108613965	12/2017	CN	G01N 21/65
113341037	12/2020	CN	N/A

OTHER PUBLICATIONS

EPO Machine Translation of CN108152385A (Year: 2025). cited by examiner

EPO Machine Translation of CN 108613965A (Year: 2025). cited by examiner

Primary Examiner: Wecker; Jennifer

Assistant Examiner: Castaneda; Steven Ray

Attorney, Agent or Firm: HAUPTMAN HAM, LLP

Background/Summary

RELATED APPLICATIONS (1) The present application is a U.S. Continuation Application of International Application NO. PCT/CN2021/143234 filed Dec. 30, 2021, and claims priority to

INCORPORATION BY REFERENCE

(1) sequence listing provided in the file entitled C6160-022_SQL_v2.xml, which is an Extensible Markup Language (XML) file that was created on Nov. 13, 2022, and which comprises 66,298 bytes, is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

(2) The present disclosure relates to the technical field of food detection, and in particular to a method for detecting the contents of whey protein and casein, and/or the ratio thereof in milk powder by liquid chromatography-mass spectrometry.

BACKGROUND ART

(3) Whey protein, as one of the main nutrients in infant formula milk powder, accounts for a particularly important nutritional proportion of the formula milk powder. The main ingredients of whey protein are shown in the following table:

(4) TABLE-US-00001 TABLE 1 Main ingredients of whey protein Relative Whey Proportion
molecular Isoelectric protein (%) mass point Function α -La 20~25% 14.2 4.20-4.50 binding to metal ions including calcium; contributing to the transportation and absorption of calcium; participating in the synthesis of lactose; regulating neurotransmitters to help sleep; promoting infant brain development; having anti-cancer effect. β -Lg 50-55% 18.3 5.35-5.49 binding to fat-soluble vitamins and fatty acids and transporting them; participating in the synthesis of prostaglandin-related enzymes, and having antihypertensive and antitumor effects. Ig 10-15% <150.0 5.50-8.30 resisting to microbial pathogens and toxins, and being one important component of human immunity; being able to protect breast from infection. BSA 5-10% 66.4 5.13 participating in the transport, metabolism, and distribution of ligands; regulating blood osmotic pressure.

(5) Whey protein contains a variety of essential amino acids required by human body and exhibits high bioavailability. It contains one third of branched-chain amino acids (leucine, isoleucine, and valine, etc.), which not only provide energy for the body, but also have been proved to be one of the important precursors for the synthesis of glutamate in human body. In addition, leucine can also participate in the intracellular metabolic pathways regulating synthesis of skeletal muscle protein, thus enhancing the synthesis of skeletal muscle protein.

(6) Whey protein is also an important source of sulfur-containing amino acids (cysteine) in human body, which are important precursors for making up glutathione and taurine. Glutathione plays a key role in free radical scavenging and immune response. Glutathione, as well as taurine, can inhibit lipid peroxidation and play an antioxidant role. Threonine in whey protein is converted directly into intestinal mucin after being taken up by intestinal tract, thus providing protection for intestinal cells as well as intestinal barrier integrity. Finally, lysine and arginine in whey protein can promote muscle growth by stimulating secretion of metabolic hormones in the body.

(7) As an important detection index for formula milk powder, it is stipulated by the national food safety standard GB 10765-2010 that the content of whey protein in infant formula milk powder should exceed 60%. Due to the complex additives in infant formula milk powder and the heating process in the production thereof, changes in the space structures of whey and casein proteins may occur, for example, a disulfide bond unfolds, resulting in protein aggregation, and thus forming polymers of various proteins, which makes the detection of whey protein in formula milk powder extremely difficult.

(8) At present, many methods have been applied to detect the content of whey protein. However, among these methods, sodium dodecyl sulfate-polyacrylamide gel electrophoresis, high performance liquid chromatography, high performance capillary electrophoresis, etc., are all affected by protein denaturation occurred in the production thereof, resulting in an inaccurate

quantification of whey protein, i.e., a large deviation in the measurement results; the amino acid conversion method, which is usually used to determine the content of whey protein, has a narrow application range and is easily affected by additives; and the liquid chromatography detection method using isotope internal standard cannot be widely used in practical production due to the need to synthesize internal standard isotopes which produces high detection cost.

(9) Casein protein is one of the main proteins in milk and can be hydrolyzed into a variety of bioactive peptides after being digested in gastrointestinal tract. At present, detection of casein protein in formula milk powder is mainly performed by high performance liquid chromatography, in which, however, casein protein shows a less desirable peak shape in high performance liquid phase, and it is difficult to distinguish the chromatographic peak of κ -casein from that of a S2-casein. Enzyme-linked immunosorbent assay (ELISA) and liquid chromatography-mass spectrometry (LC-MS) and the like have also been proposed in succession, and they show a good performance in the detection of casein protein.

(10) The information disclosed in the Background section is only intended to facilitate understanding of the general background of the present disclosure and should not be taken as an acknowledgment or implying, in any form, that the information constitutes the prior art already known to a person with ordinary skill in the art.

SUMMARY OF THE INVENTION

Purpose

(11) In view of the disadvantages of the current detection methods as mentioned above, the present disclosure aims to provide a method for detecting the contents of whey protein and casein, and/or the ratio thereof in milk powder by liquid chromatography-mass spectrometry. According to the method of the present disclosure, the accuracy and stability of detecting peptide segments by mass spectrometry using external standard scan can be improved by selecting the respective characteristic peptide segments of whey protein and casein and using the same for correction.

Solutions

(12) In order to achieve the purpose of the invention, the technical solutions provided by the present disclosure are as follows.

(13) In one aspect, provided is a method for detecting the contents of whey protein and casein, and/or the ratio thereof in milk powder by liquid chromatography-mass spectrometry, which comprises: 1) preparing a standard solution of whey protein with the characteristic peptide segments of α -lactalbumin and β -lactoglobulin, in accordance with the α -lactalbumin and β -lactoglobulin accounting for 20% and 50% of the whey protein, respectively; 2) preparing a standard solution of casein protein with the characteristic peptide segments of α -casein and β -casein, in accordance with the α -casein and β -casein accounting for 50% and 40% of the casein protein, respectively; 3) mixing the standard solution of whey protein from step 1) with the standard solution of casein protein from step 2) to prepare a series of mixed protein standard solutions with a series of whey protein:casein ratios; 4) drawing standard curves with the ratios of whey protein to casein as abscissa and the peak area ratios of characteristic peptide segments of α -lactalbumin to β -casein and those of β -lactoglobulin to α -casein as ordinate; 5) detecting the milk powder to be tested for the peak area ratios of characteristic peptide segments of β -lactoglobulin to α -casein and those of α -lactalbumin to β -casein; 6) substituting the peak area ratio of the characteristic peptide segments of β -lactoglobulin to α -casein in the detected milk powder into the standard curve to determine the ratio M of whey protein to casein, and calculate the actual amounts of β -lactoglobulin and α -casein based on the ratio M thus determined; and, substituting the peak area ratio of the characteristic peptide segments of α -lactalbumin to β -casein in the detected milk powder into the standard curve to determine the ratio N of whey protein to casein, and calculate the actual amounts of α -lactalbumin to β -casein based on the ratio N thus determined; 7) acquiring the actual contents of whey protein and casein, and/or the actual ratio of whey protein/casein based on the actual amounts of α -lactalbumin, β -casein, β -lactoglobulin, and α -casein obtained from step 6).

(14) In the above method, preferably, the characteristic peptide segments of α -lactalbumin, β -casein, β -lactoglobulin, and α -casein are LDQWLCEK shown in SEQ ID NO. 1, VLPVPQK shown in SEQ ID NO. 2, TPEVDDEALEK shown in SEQ ID NO. 3, and FALPQYLK shown in SEQ ID NO. 4, respectively.

(15) Preferably, in step 3), the series of whey protein:casein ratios include whey protein:casein ratios of 0.25, 0.43, 0.67, 1, 1.5, and 2.33, respectively.

(16) Preferably, in step 6), the actual amounts of β -lactoglobulin and α -casein are calculated by the following formulas, respectively:

(17) β -lactoglobulin = $\frac{M1}{M1+M2}X * 50\%$, α -casein = $\frac{M2}{M1+M2}X * 50\%$; wherein, M1 is the preceding term in the ratio M of whey protein:casein as determined based on the peak area ratio of characteristic peptide segments of β -lactoglobulin to α -casein, and M2 is the latter term in the ratio M of whey protein:casein as determined based on the peak area ratio of characteristic peptide segments of β -lactoglobulin to α -casein, with the provisions of: $M=M1/M2$, and $M1+M2=1$; and, wherein, X is the total protein concentration of the sample.

(18) Preferably, in step 6), the actual amounts of α -lactalbumin and β -casein are calculated by the following formulas:

(19) α -la = $\frac{N1}{N1+N2}X * 20\%$, β -cs = $\frac{N2}{N1+N2}X * 40\%$; wherein, N1 is the preceding term in the ratio N of whey protein:casein as determined based on the peak area ratio of characteristic peptide segments of α -lactalbumin to β -casein, and N2 is the latter term in the ratio N of whey protein:casein as determined based on the peak area ratio of characteristic peptide segments of α -lactalbumin to β -casein, with the provisions of: $N=N1/N2$, and $N1+N2=1$; and, wherein, X is the total protein concentration of the sample.

(20) Preferably, in step 7), the formula for calculating the actual content of whey protein is:

(21) $\frac{\frac{N1}{N1+N2}X * 20\% + \frac{M1}{M1+M2}X * 50\%}{0.7}$ the formula for calculating the actual content of casein is:

(22) $\frac{\frac{M2}{M1+M2}X * 50\% + \frac{N2}{N1+N2}X * 40\%}{0.9}$ the formula for calculating the actual ratio of whey protein/casein is:

(23) $\frac{\frac{N1}{N1+N2}X * 20\% + \frac{M1}{M1+M2}X * 50\%}{0.7} / \frac{\frac{M2}{M1+M2}X * 50\% + \frac{N2}{N1+N2}X * 40\%}{0.9} = \frac{18N1 + 45M1}{35M2 + 28N2} = \frac{63MN + 18N + 45M}{35N + 28M + 63}$; wherein, M1 is the preceding term in the ratio M of whey protein:casein as determined based on the peak area ratio of characteristic peptide segments of β -lactoglobulin to α -casein, and M2 is the latter term in the ratio M of whey protein:casein as determined based on the peak area ratio of characteristic peptide segments of β -lactoglobulin to α -casein, with the provisions of: $M=M1/M2$, and $M1+M2=1$; wherein, N1 is the preceding term in the ratio N of whey protein:casein as determined based on the peak area ratio of characteristic peptide segments of α -lactalbumin to β -casein, and N2 is the latter term in the ratio N of whey protein:casein as determined based on the peak area ratio of characteristic peptide segments of α -lactalbumin to β -casein, with the provisions of: $N=N1/N2$, and $N1+N2=1$; and, wherein, X is the total protein concentration of the sample.

(24) Preferably, the detection in step 5) comprises the steps of sample treatment and enzyme digestion: dissolving the milk powder sample to be tested into a sample solution with a protein concentration of 0.1-0.4 mg/mL, preferably 0.2 mg/mL; subjecting the sample solution to rough filtration with a pure acetic acid filter membrane preferably with a pore size of 0.45 μ m; adding the roughly filtered sample solution into ammonium bicarbonate solution, followed by adding dithiothreitol solution, and standing in a water bath at 65-75° C., preferably 70° C. for 25-35 min, preferably 30 min; after cooling, adding iodoacetamide solution, and standing for 25-35 min, preferably 30 min in the dark; illuminating the mixture for 8-12 min, preferably 10 min, followed by adding calcium chloride solution; then adding trypsin solution to allow an enzyme digestion at 37° C. for 26-30 h; then adding acetic acid solution to stop the enzyme digestion; filtering the reaction mixture with polyethersulfone filter membrane, and then performing selective ion scanning analysis by mass spectrometry.

(25) In a specific embodiment, the upper detection limit of the method of the present disclosure is at a total protein concentration of 0.4 mg/mL.

(26) In a second aspect, provided is a combination of characteristic peptide segments for detecting the ratio of whey protein:casein by liquid chromatography-mass spectrometry, comprising: characteristic peptide segments of α -lactalbumin, β -casein, β -lactoglobulin, and α -casein with amino acid sequences as shown in SEQ ID NO. 1, SEQ ID NO. 2, SEQ ID NO. 3, and SEQ ID NO. 4, respectively.

(27) In a specific embodiment, the combination of characteristic peptide segments consists of characteristic peptide segments of α -lactalbumin, β -casein, β -lactoglobulin, and α -casein with amino acid sequences as shown in SEQ ID NO. 1, SEQ ID NO. 2, SEQ ID NO. 3 and SEQ ID NO. 4, respectively.

Beneficial Effects

(28) Compared with the prior art, the relative quantitative method for detecting the contents of whey protein and casein, and/or the ratio thereof in milk powder has high accuracy and stability, and in addition, it has the following advantages: easy to pretreat samples, low cost, a wide range of applicability, enable to perform quantitative detection of whey protein at one time, not affected by protein denaturation in the production process, no need to synthesize isotope internal standard, and avoiding influence of ionization efficiency and matrix in mass spectrometry detection by means of their own multiple characteristic peptide segment ratios.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) One or more examples are exemplified by the pictures in the accompanying drawings that correspond thereto and are not intended to be limiting of the embodiments. As used herein, the word “exemplary” means “serving as an example, embodiment, or illustrative”. Any embodiment described herein as “exemplary” is not necessarily to be construed as superior or better than other embodiments.

(2) FIG. 1 is a full scanning mass spectrum of the mixed standard substance used in the Example;

(3) FIG. 2 is a full scanning mass spectrum of the milk powder sample;

(4) FIG. 3 is the mass spectrum of four characteristic peptide segments in PRM mode;

(5) FIG. 4 is the full fragment ion diagram of the characteristic peptide segment (LDQWLCEK) shown in SEQ ID NO. 1;

(6) FIG. 5 is the full fragment ion diagram of the characteristic peptide segment (TPEVDDEALEK) shown in SEQ ID NO. 3;

(7) FIG. 6 is the full fragment ion diagram of the characteristic peptide segment (FALPQYLK) shown in SEQ ID NO. 4;

(8) FIG. 7 is the full fragment ion diagram of the characteristic peptide segment (VLPVPQK) shown in SEQ ID NO. 2;

(9) FIG. 8 is a quantitative daughter ion peak profile of four characteristic peptide segments;

(10) FIG. 9 shows the change of peak area ratio of characteristic peptide segments LDQWLCEK (SEQ ID NO. 1)/VLPVPQK (SEQ ID NO. 2) at different enzyme concentrations and with different enzymolysis time, with the enzymolysis time as abscissa, and the peak area ratio of two characteristic peptide segments as ordinate;

(11) FIG. 10 shows the change of peak area ratio of characteristic peptide segments TPEVDDEALEK (SEQ ID NO. 3)/FALPQYLK (SEQ ID NO. 4) at different enzyme concentrations and with different enzymolysis time, with the enzymolysis time as abscissa, and the peak area ratio of two characteristic peptide segments as ordinate;

(12) FIG. 11 shows the impact of different substrate concentrations on the peak area ratio of characteristic peptide segments LDQWLCEK (SEQ ID NO. 1)/VLPVPQK (SEQ ID NO. 2) at an enzyme concentration of 1:20 with enzymolysis time of 28 h, with the substrate concentration as

- abscissa, and the peak area ratio of two characteristic peptide segments as ordinate;
- (13) FIG. 12 shows the impact of different substrate concentrations on the peak area ratio of characteristic peptide segments TPEVDDEALEK (SEQ ID NO. 3)/FALPQYLK (SEQ ID NO. 4) at an enzyme concentration of 1:20 with enzymolysis time of 28 h, with the substrate concentration as abscissa, and the peak area ratio of two characteristic peptide segments as ordinate;
- (14) FIG. 13 is a standard curve of peak area ratio vs concentration ratio of characteristic peptide segments TPEVDDEALEK (SEQ ID NO. 3)/FALPQYLK (SEQ ID NO. 4), with the concentration ratio of two characteristic peptide segments as abscissa, and the peak area ratio thereof as ordinate; and R2 of the standard curve regression equation is 0.9935;
- (15) FIG. 14 is a standard curve of peak area ratio vs concentration ratio of characteristic peptide segments LDQWLCEK (SEQ ID NO. 1)/VLPVPQK (SEQ ID NO. 2), with the concentration ratio of two characteristic peptide segments as abscissa, and the peak area ratio thereof as ordinate; and R2 of the standard curve regression equation is 0.992;
- (16) FIG. 15 is a primary mass spectrum of desalted whey powder; and
- (17) FIG. 16 shows the composition of desalted whey powder protein.

DETAILED DESCRIPTION OF THE INVENTION

- (18) In order to make the purpose, technical solutions, and advantages of the embodiments of the invention clearer, the technical solutions in the embodiments of the present disclosure will be described clearly and completely, obviously, the described embodiments are some of the embodiments of the present disclosure, but not all of them. Based on the embodiments of the present disclosure, all other embodiments obtained by those of ordinary skill in the art without creative work are within the scope of the present invention.
- (19) In addition, in order to better explain the present invention, a lot of specific details are given in the following embodiments. It will be understood by those skilled in the art that the present invention may be practiced without certain specific details. In some embodiments, materials, elements, methods, means, etc., well known to those skilled in the art, are not described in detail so as to highlight the spirit of the present invention.
- (20) Throughout the specification and claims, the term “comprising” or variations thereof, such as “including” or “containing” and the like, will be understood to include the stated components and not to exclude other elements or other components, unless expressly indicated otherwise.
- (21) In the following examples, the reagents used were as follows:
- (22) Dithiothreitol (sigma, $\geq 98\%$ HPLC); iodoacetamide (sigma, $\geq 99\%$ NMR); α -lactalbumin standard substance (sigma, $\geq 85\%$); β -lactoglobulin standard substance (sigma, $\geq 90\%$); α -casein standard substance (sigma, $\geq 70\%$); β -casein standard substance (sigma, $\geq 98\%$); bovine basic trypsin (sigma, $\geq 10,000$ BAEE); Calcium chloride (AR); Ammonium bicarbonate (AR); Acetic acid (AR).
- (23) In the following examples, nano-high-performance liquid chromatography-orbitrap high resolution mass spectrometry (NanoLC-Orbitrap MS) was used for selective ion scanning analysis under the following chromatographic conditions: Nano-high-performance liquid UltiMate 3000 (ACCELA) Column: Acclaim® PepMap RSLC (75 μm ×15 cm, nano Viper C18, 2 μm , 100 Å Thermo Fisher Scientific); Acclaim PepMap™ 100 (75 μm ×2 cm, nanoViper C18, 3 μm , 100 Å Thermo Fisher Scientific) Column temperature: 50° C. Mobile phase A: 2% acetonitrile+98% water+0.1% formic acid Mobile phase B: 80% acetonitrile+20% water+0.1% formic acid Loading: 2% acetonitrile+98% water Flow rate: NC: 0.25 $\mu\text{L}/\text{min}$, loading flow rate: 0.3 $\mu\text{L}/\text{min}$ Injection volume: 1 μL Gradient elution conditions Gradient elution
- (24) TABLE-US-00002 Elution time (min) Flow rate ($\mu\text{L}/\text{min}$) A % B % 0 0.25 96 4 5 0.25 96 4 65 0.25 78 22 70 0.25 10 90 75 0.25 96 4 76 0.25 96 4
- (25) In the following examples, the characteristic peptide segments were subjected to a mass spectrometry analysis in PRM mode under the following mass spectrometry conditions: Q Exactive mass spectrometer (Thermo Fisher Scientific) PRM mode Running time: 5-75 min Polarity:

positive Resolution: 17500 AGC target: 1e5 Maximum IT: 100 ms Separation window: 1.0 m/z
Initial mass-to-charge ratio: Fixed first mass NCE: 27%, 28%.

Example 1: Selection of Characteristic Peptide Segments

(26) A mixed standard substance of α -lactalbumin, β -lactoglobulin, α -casein, and β -casein and a milk powder sample were subjected to a full-scanning by mass spectrometry, and the mass spectra thereof were shown in FIG. 1 and FIG. 2, respectively.

(27) By using ExPASyPeptideMass website, four proteins obtained by retrieval in NCBI's database, α -lactalbumin, β -lactoglobulin, α -casein, and β -casein, were subjected to a simulated enzyme digestion (for the peptide segments produced by the simulated digestion, please see Table 2 below), and a comparison. The compared peptide segments were substituted to NCBI website for BLAST alignment analysis, and peptide segments with high response value and a sequence length of 5-25 amino acids were selected and finally determined as the specific peptide segments, which were: characteristic peptide segment of α -lactalbumin (hereinafter also referred to as α -la) LDQWLCEK (SEQ ID NO. 1), characteristic peptide segment of β -lactoglobulin (hereinafter also referred to as β -lg) TPEVDDEALEK (SEQ ID NO. 3), characteristic peptide segment of α -casein (hereinafter also referred to as α -cs) FALPQYLK (SEQ ID NO. 4), characteristic peptide segment of β -casein (hereinafter also referred to as β -cs) VLPVPQK (SEQ ID NO. 2).

(28) Specific information on the above four characteristic peptide segments was shown in Table 3, and their mass spectra in PRM mode were shown in FIG. 3.

(29) The respective daughter ion information maps of the above four characteristic peptide segments were shown in FIGS. 4, 5, 6, and 7, respectively, from which it can be seen that the quantitative daughter ions selected in the experiment have higher response values in the detection process; FIG. 8 showed the daughter ion peaks of four peptide segments detected separately (here, they were combined into one diagram), to illustrate that they were independent peaks and did not interfere with each other during detection; in addition, the quantitative fragment ion information was shown in Table 4; and fragment ions with appropriate response value and mass number of each specific peptide segment were selected as quantitative ions.

(30) Table 2.—Peptide segments produced by simulated digestion of α -lactalbumin, β -lactoglobulin, α -casein, and β -casein (SEQ ID NOS. 5-75 below)

(31) β -Lactoglobulin

(32) Protein sequence (SEQ ID NO. 5):

(33) TABLE-US-00003 LIVTQTMKGL DIQKVAGTWY SLAMAASDIS
LLDAQSAPLR VYVEELKPTP EGDLEILLQKWENDECAQKK IIAEKT KIPA
VFKIDALNEN KVLVLDTDYK KYLLFCMENS AEPEQSLVCQCLVRTPEVDD
EALEKFDKAL KALPMHIRLS FNPTQLEEQC HI

(34) Peptide fragments by theoretical digestion (as shown below in SEQ ID NOS. 6-18, respectively):

(35) TABLE-US-00004 Arti- Posi- # ficial Peptide Mass tion MC Mod(s) Mods Sequence
2707.3759 15-40 0 VAGTWYSLA MAASDISLL DAQSAPLR (SEQ ID NO. 6) 2675.2336
102-124 0 Cys_CAM: 2846.2980 YLLFCMENS 106, 119 AEPEQSLVC 121 QCLVR (SEQ ID
NO. 7) 2313.2587 41-60 0 VYVEELKPT PEGDLEILL QK (SEQ ID NO. 8) 1658.7843
149-162 0 Cys_CAM: 1715.8057 LSFNPTQLE 160 EQCHI (SEQ ID NO. 9) 1245.5845 125-
135 0 TPEVDDEAL EK (SEQ ID NO. 10) 1122.4520 61 69 0 Cys_CAM: 1179.4735
WENDECAQK 66 (SEQ ID NO. 11) 1065.5826 92-100 0 VLVLDTDYK (SEQ ID NO.
12) 933.5437 1-8 0 LIVTQTMK (SEQ ID NO. 13) 916.4734 84-91 0 IDALNENK
(SEQ ID NO. 14) 837.4763 142-148 0 ALPMHIR (SEQ ID NO. 15) 674.4235 78-
83 0 IPAVFK (SEQ ID NO. 16) 673.3879 9-14 0 GLDIQK (SEQ ID NO. 17)
573.3606 71-75 0 IIAEK (SEQ ID NO. 18)

F Chain of Bovine α -Lactalbumin, Crystal Structure

(36) Protein sequence (SEQ ID NO. 19):

(37) TABLE-US-00005 EQLTKCEVFR ELKDLKGYGG VSLPEWVCTT
FHTSGYDTQA IVQNNNDSTEY GLFQINNKIWCKDDQNPHSS NICNISCDF
LDDDLTDDIM CVKKILDKVG INYWLAHKAL CSEKLDQWLCEKL

(38) Peptide fragments by theoretical digestion (as shown below in SEQ ID NOS. 20-28, respectively):

(39) TABLE-US-00006 Arti- Posi- # ficial Peptide Mass tion MC Mod(s) Mods Sequence
4654.1467 17-58 0 Cys_CAM: 4711.1681 GYGGVSLPE 28 WVCTTFHTS GYDTQAIV
QNNNDSTEYGLFQINN (SEQ ID NO. 20) 1889.7752 63-79 0 Cys CAM: 2003.8182
DDQNPHSSN 73, 77 ICNISCDF (SEQ ID NO. 21) 1642.7338 80-93 0 Cys_CAM:
1699.7553 FLDDDLTDD 91 IMCVK (SEQ ID NO. 22) 1200.6524 99-108 0 VGINYWLAH
K (SEQ ID NO. 23) 1034.4975 115-122 0 Cys_CAM: 1091.5190 LDQWLCEK 120
(SEQ ID NO. 24) 653.3075 6-10 0 Cys_CAM: 710.3290 CEVFR 6 (SEQ ID
NO. 25) 650.3178 109-114 0 Cys_CAM: 707.3392 ALCSEK 111 (SEQ ID NO. 26)
618.3457 1-5 0 EQLTK (SEQ ID NO. 27) 549.2853 59-62 0 Cys_CAM: 606.3068
IWCK 61 (SEQ ID NO. 28)

α -S1-Casein, Partial [Bovine]

(40) Protein sequence (SEQ ID NO. 29):

(41) TABLE-US-00007 MKLLILTCLV AVALARPKHP IKHQGLPQEV LNENLLRFFV
APFPEVFGKE KVNELSKDIGSESTEDQAME DIKQMEAESI SSSEEIVPNS
VEQKHIQKED VPSEYLGYLEQLRLKKYKVPQLEIVPNS AEERLHSMKE
GIHAQQKEPM IGVNQELAYF YPE

(42) Peptide fragments by theoretical digestion (as shown below in SEQ ID NOS. 30-42, respectively):

(43) TABLE-US-00008 Arti- Posi- # ficial Peptide Mass tion MC Mod(s) Mods Sequence
2321.0813 74-94 0 QMEAESS SEEIVPNSV EQK (SEQ ID NO. 30) 1899.8833 148-
163 0 EPMIGVNQE LAYFYPE (SEQ ID NO. 31) 1767.7589 58-73 0 DIGSESTED
QAMEDIK (SEQ ID NO. 32) 1759.9449 23-37 0 HQGLPQEV LNENLLR (SEQ ID
NO. 33) 1694.0760 3-18 0 Cys_CAM: 1751.0975 LLILTCLVA 8 VALARPK (SEQ
ID NO. 34) 1580.8278 121-134 0 VPQLEIVPN SAEER (SEQ ID NO. 35) 1334.7299 38-
49 0 FFVAPFPEV FGK (SEQ ID NO. 36) 1267.7045 106-115 0 YLGYLEQLLR (SEQ ID
NO. 37) 910.4741 140-147 0 EGIHAQQK (SEQ ID NO. 38) 831.3843 99-105 0
EDVPSE (SEQ ID NO. 39) 689.3828 52-57 0 VNELSK (SEQ ID NO. 40)
615.3283 135-139 0 LHSMK (SEQ ID NO. 41) 525.3143 95-98 0 HIQK (SEQ ID
NO. 42)

α -S2-Casein Precursor [Bovine]

(44) Protein sequence (SEQ ID NO. 43):

(45) TABLE-US-00009 MKFFIFTCLL AVALAKNTME HVSSSEESII SQETYKQEK
MAINPSKENL CSTFCKEVVRNANEEYSIG SSSEESA EVA TEEVKITVDD
KHYQKALNEI NQFYQKFPQY LQYLYQGPIVLNPWDQVKRN AVPTPTLNR
EQLSTSEENS KKTVDMESTE VFTKLTKLTE EEKNRLNFLKKISQRYQKFA
LPQYLKTVYQ HQKAMKPWIK PKTKVIPYVR YL

(46) Peptide fragments by theoretical digestion (as shown below as SEQ ID NOS. 44-63, respectively):

(47) TABLE-US-00010 Arti- Posi- # ficial Peptide Mass tion MC Mod(s) Mods Sequence
2709.4075 107-128 0 FPQYLQYLY QGPIVLNPW DQVK (SEQ ID NO. 44) 2688.1642
61-85 0 NANEEYSI GSSSEESAE VATEEVK (SEQ ID NO. 45) 2299.0394 17-36 0
NTMEHVSSS EESIISQET YK (SEQ ID NO. 46) 1556.8909 3-16 0 Cys_CAM:
1613 9123 FFIFTCLLA 8 VALAK (SEQ ID NO. 47) 1386.6457 153-164 0
TVDMESTEV FTK (SEQ ID NO. 48) 1367.6954 96-106 0 ALNEINQFY QK (SEQ
ID NO. 49) 1251.5699 141-151 0 EQLSTSEEN SK (SEQ ID NO. 50) 1195.6793 130-140

NAVITPTL NR (SEQ ID NO. 51) 1098.6128 204-212 0 AMKPWIQPK (SEQ ID NO. 52) 1044.4489 48-56 0 Cys_CAM: 1158.4918 ENLCSTFCK 51, 55 (SEQ ID NO. 53) 979.5611 189-196 0 FALPOYLK (SEQ ID NO. 54) 903.4683 197-203 0 TVYQHQQK (SEQ ID NO. 55) 874.4451 40-47 0 NMAINPSK (SEQ ID NO. 56) 748.3723 168-173 0 LTEEEK (SEQ ID NO. 57) 746.4559 215-220 0 VIPYVR (SEQ ID NO. 58) 690.3668 86-91 0 ITVDDK (SEQ ID NO. 59) 634.3922 176-180 0 LNFLK (SEQ ID NO. 60) 575.2936 92-95 0 HYQK (SEQ ID NO. 61) 503.2936 182-185 0 ISQR (SEQ ID NO. 62) 502.2983 57-60 0 EVVR (SEQ ID NO. 63)

β -Casein

(48) Protein sequence (SEQ ID NO. 64):

(49) TABLE-US-00011 MKVLILACLV ALALARELEE LNVPGEIVES LSSSEESITR INKKIEKFQS EEQQQTEDELQDKIHPPFAQT QSLVYPFPGP IHNSLPQNIP PLTQTPVVVP PFLQPEVMGV SKVKEAMAPKHKEMPFPKYP VEPFTESQSL TLTDVENLHL PLLLQSWNH QPHQPLPPTV MFPPQSVLSLSQSKVLPYPQ KAVPYPQRDM PIQAFLLYQE PVLGPVRGPF PIIV

(50) Peptide fragments by theoretical digestion (as shown below in SEQ ID NOS. 65-75, respectively):

(51) TABLE-US-00012

Arti-Sequence	Posi- #	ficial Peptide	Mass	tion MC	Mod(s)	Mods
6359.2558 129-184 0	YPVEPFTES	OSLTLTDVE	NLHLPLPLL	QSWMHQPHQ		
PLPPTVMFP PQSVLSLSQ SK (SEQ ID NO. 65)	5356.8594	64-112 0	IHPFAQTQS			
LVYPPGPI HNSLPQNIP PLTQTPVWP PFLQPEVMG VSK (SEQ ID NO. 66)	2646.2992	17-40 0	ELEELNVPG EIVESLSSS	EESITR (SEQ ID NO. 67)	2186.1678	199-217 0
DMPIQAFLL YQEPVLGPV R (SEQ ID NO. 68)	1981.8621	48-63 0	FQSEEQQQT			
EDELQDK (SEQ ID NO. 69)	1438 9177	3-16 0	VLILACLVA LALAR (SEQ ID NO. 70)	830.4519	192-198 0	Cys_CAM8 1751.0975
AVPYPQR (SEQ ID NO. 71)	780.4978	185-191 0	VIPVPQK (SEQ ID NO. 72)	748.3698	123-128 0	EMPFPK
(SEQ ID NO. 73)	742.4497	218-224 0	GPFPIIV (SEQ ID NO. 74)	646.3228	115-120 0	EAMAPK (SEQ ID NO. 75)

(52) TABLE-US-00013

TABLE 3	Information on characteristic peptide segments	Protein	Peptide Molecular weight	Amino acid chain	β -lg	TPEVDDEALEK	623.29	Thr-Pro-Glu-Val-Asp-	(SEQ ID NO. 3)	Lys α -la	LDQWLCEK	546.23
Leu-Asp-Gln-Trp-Leu-	(SEQ ID NO. 1)	α -cs	FALPQYLK	490.19	Phe-Ala-Leu-Pro-Gln-	(SEQ ID NO. 4)	β -cs	VLPVPQK	390.75	Val-Leu-Pro-Val-Pro-	(SEQ ID NO. 2)	

(53) TABLE-US-00014

TABLE 4	Information on quantitative fragment ions	Daughter Retention	Parent ion mass	time	particle number	(KDa)	(min)	TPEVDDEALEK	199.10836	27.5	
(SEQ ID NO. 3)	LDQWLCEK	268.16794	55.2	(SEQ ID NO. 1)	FALPQYLK	120.08216	57.8	(SEQ ID NO. 4)	VLPVPQK	372.2257	23.7
(SEQ ID NO. 2)				(SEQ ID NO. 2)							

Example 2: Optimization of Enzymolysis Conditions, Selection of Substrate Concentration, as Well as Selection of Filter Membrane

(54) Optimization of Enzymolysis Conditions:

(55) In this experiment, milk powder samples under the brand names Sanyuan and Ailiyou (with a protein concentration of 11.6 g/100 g milk powder sample) were used as substrates and subjected to enzymolysis at enzyme concentrations of 1:20, 1:40, 1:60, 1:80, and 1:100, respectively, according to the information provided by the product suppliers. As a result, an enzyme concentration of 1:20 and enzymolysis time of 28 h were selected based on variations in the peak area ratio of the characteristic peptide segments LDQWLCEK (SEQ ID NO. 1)/VLPVPQK (SEQ ID NO. 2) and variations in the peak area ratio of the characteristic peptide segments TPEVDDEALEK (SEQ ID NO. 3)/FALPQYLK (SEQ ID NO. 4) in the process of enzymolysis, both of which tended to be

stable after enzymolysis for 28 h at the enzyme concentration of 1:20 (See FIGS. 9 and 10 for details).

(56) Selection of Substrate Concentration:

(57) Milk powder samples were dissolved into solutions with ten protein concentration gradients between 0.1-1.0 mg/mL, which were then filtered roughly with pure acetic acid filter membrane with a pore size of 0.45 μ m. 250 μ L of sample solution was added to 150 μ L of ammonium bicarbonate solution, followed by adding 10 μ L of dithiothreitol solution, and kept in a water bath at 70° C. for 30 min. After cooling, 30 μ L of iodoacetamide solution was added thereto, followed by standing for 30 min in the dark. After illuminating for 10 min, 10 μ L of calcium chloride solution was added, followed by adding 50 μ L of Trypsin solution, and then enzyme digestion was performed at 37° C. for 28 h, followed by adding 10 μ L of acetic acid solution, and standing for 15 min to stop digestion. After filtration with 0.22 μ m of polyethersulfone filter membrane, selective ion scanning analysis was carried out by nanoliter liquid chromatography. The results showed that when the protein concentration was below 0.4 mg/mL, the ratio of two characteristic peptide segments remained relatively stable after enzymolysis, indicating a good detection effect (See FIGS. 11 and 12).

(58) Selection of Filter Membrane:

(59) The pure acetic acid filter membrane having a large pore size of 0.45 μ m was used to filter the milk powder sample preliminarily, to remove some additives, so as to avoid the impact thereof on the subsequent experiments and ensure the stability of the method. 0.22 μ m of polyethersulfone filter membrane with extremely low protein adsorption was selected for filtration before running on machine.

Example 3: Detection Procedure of the Method of the Present Disclosure

(60) 1) Establishment of Standard Curve:

(61) Four protein standard substances of α -lactalbumin, β -lactoglobulin, α -casein, and β -casein were used. Specifically, whey protein standard solution was prepared in accordance with α -lactalbumin and β -lactoglobulin accounting for 20% and 50% of whey protein, respectively (i.e., the contents of α -lactalbumin and β -lactoglobulin were 20% and 50%, respectively); and casein standard solution was prepared in accordance with α -casein and β -casein accounting for 50% and 40% of casein, respectively (i.e., the contents of α -casein and β -casein were 50% and 40%, respectively); and the prepared whey protein standard solution had the same concentration as the prepared casein standard solution.

(62) Certain amounts of the whey protein standard solution and the casein standard solution prepared as above were mixed to prepare 1 mL of a mixed solution with a whey protein/casein ratio of 0.25, 0.43, 0.67, 1, 1.5, 2.33 (mass ratio), which was then detected for drawing standard curves; the standard curve of peak area ratio vs concentration ratio of characteristic peptide segments TPEVDDEALEK (SEQ ID NO. 3)/FALPQYLK (SEQ ID NO. 4) was shown in FIG. 13, and the standard curve of peak area ratio vs concentration ratio of characteristic peptide segments LDQWLCEK (SEQ ID NO. 1)/VLPVPQK (SEQ ID NO. 2) was shown in FIG. 14.

(63) 2) Treatment and Enzyme Digestion of Milk Powder Samples to be Tested:

(64) Dissolving the milk powder sample to be tested into a sample solution with a protein concentration of 0.2 mg/ml; subjecting the sample solution to rough filtration with a pure acetic acid filter membrane with a pore size of 0.45 μ m; adding the roughly filtered sample solution into ammonium bicarbonate solution, followed by adding dithiothreitol solution, and placing the mixture in a water bath at 70° C. for 30 min; after cooling, adding iodoacetamide solution thereto, and standing for 30 min in the dark; then illuminating for 10 min, followed by adding calcium chloride solution thereto; adding trypsin solution to allow an enzyme digestion at 37° C. for 26-30 h; adding acetic acid solution to stop the enzyme digestion; 3) filtering the enzymolysis products from step 2) with polyethersulfone filter membrane, and then performing selective ion scanning analysis by mass spectrometry to detect the peak area ratios of characteristic peptide segments of β -

lactoglobulin to α -casein and of α -lactalbumin to β -casein in the milk powder; 4) substituting the peak area ratio of the characteristic peptide segments of β -lactoglobulin to α -casein in the milk powder as detected from step 3) into the standard curve to determine the ratio M of whey protein to casein, and calculating the actual amounts of β -lactoglobulin and α -casein based on the ratio M thus determined; and, substituting the peak area ratio of the characteristic peptide segments of α -lactalbumin to β -casein in the milk powder as detected from step 3) into the standard curve to determine the ratio N of whey protein to casein, and calculating the actual amounts of α -lactalbumin to β -casein based on the ratio N thus determined; the actual amounts of β -lactoglobulin and α -casein are calculated by the following formulas, respectively:

(65) - lactoglobulin = $\frac{M_1}{M_1+M_2}X * 50\%$, - casein = $\frac{M_2}{M_1+M_2}X * 50\%$; wherein, M1 was the preceding term in the ratio M of whey protein:casein as determined based on the peak area ratio of characteristic peptide segments of β -lactoglobulin to α -casein, and M2 was the latter term in the ratio M of whey protein:casein as determined based on the peak area ratio of characteristic peptide segments of β -lactoglobulin to α -casein, with the provisions of: $M=M_1/M_2$, and $M_1+M_2=1$; and X is the total protein concentration of the sample; the actual amounts of α -lactalbumin and β -casein are calculated by the following formulas, respectively:

(66) - la = $\frac{N_1}{N_1+N_2}X * 20\%$, - cs = $\frac{N_2}{N_1+N_2}X * 40\%$; wherein, N1 was the preceding term in the ratio N of whey protein:casein as determined based on the peak area ratio of characteristic peptide segments of α -lactalbumin to β -casein, and N2 was the latter term in the ratio N of whey protein:casein as determined based on the peak area ratio of characteristic peptide segments of α -lactalbumin to β -casein, with the provisions of: $N=N_1/N_2$, and $N_1+N_2=1$; and X is the total protein concentration of the sample; 5) acquiring the actual contents of whey protein and casein, and/or the actual ratio of whey protein/casein based on the actual amounts of α -lactalbumin, β -casein, β -lactoglobulin, and α -casein obtained from step 4); the formula for calculating the actual content of whey protein was:

(67) $\frac{\frac{N_1}{N_1+N_2}X * 20\% + \frac{M_1}{M_1+M_2}X * 50\%}{0.7}$ the formula for calculating the actual content of casein was:

(68) $\frac{\frac{M_2}{M_1+M_2}X * 50\% + \frac{N_2}{N_1+N_2}X * 40\%}{0.9}$ the formula for calculating the actual ratio of whey protein/casein was:

(69) $0 \frac{\frac{N_1}{N_1+N_2}X * 20\% + \frac{M_1}{M_1+M_2}X * 50\%}{0.7} / \frac{\frac{M_2}{M_1+M_2}X * 50\% + \frac{N_2}{N_1+N_2}X * 40\%}{0.9} = \frac{18N_1 + 45M_1}{35M_2 + 28N_2} = \frac{63MN + 18N + 45M}{35N + 28M + 63}$; wherein, M1 was the preceding term in the ratio M of whey protein:casein as determined based on the peak area ratio of characteristic peptide segments of β -lactoglobulin to α -casein, and M2 was the latter term in the ratio M of whey protein:casein as determined based on the peak area ratio of characteristic peptide segments of β -lactoglobulin to α -casein, with the provisions of: $M=M_1/M_2$, and $M_1+M_2=1$; wherein, N1 was the preceding term in the ratio N of whey protein:casein as determined based on the peak area ratio of characteristic peptide segments of α -lactalbumin to β -casein, and N2 was the latter term in the ratio N of whey protein:casein as determined based on the peak area ratio of characteristic peptide segments of α -lactalbumin to β -casein, with the provisions of: $N=N_1/N_2$, and $N_1+N_2=1$; and, wherein, X is the total protein concentration of the sample.

Example 4: Methodology Validation on the Detection Method of the Present Disclosure by Using Desalted Whey Powder

(70) Standard Addition Recovery Rate

(71) Desalted whey powder was used as the standard sample and was known to have a protein content of 12% according to its product annotation, which was almost made up of whey protein, with only a very little or negligible amount of casein, as detected by an instrument; the primary mass spectrum of the desalted whey powder was shown in FIG. 15 and analyzed to obtain its composition as shown in FIG. 16.

(72) the detection method of the present disclosure was verified by using the desalted whey powder and the results were shown in Table 5 below.

(73) TABLE-US-00015 TABLE 5 Added amount Theoretical Detection Recovery RSD (mg) value

(mg) value (mg) rate (%) (%) Substrate 0 / 19.48 / 7.35 Desalted 3.60 23.08 23.03 98.63 0.84
whey 7.20 26.68 26.90 103.06 7.42 powder 10.80 30.28 31.72 113.33 1.83 14.40 33.88 34.11
101.62 0.84

(74) According to Table 5, the standard addition recovery rate of the desalted whey powder detected by the detection method of the present disclosure was 98.63%-113.33%, and the corresponding RSD was 0.84%-7.42%.

Example 5: Methodology Validation on the Detection Method of the Present Disclosure by Using Commercial Formula Milk Powder

(75) According to the detection method of the present disclosure described in Example 3, three brands of commercial formula milk powder were tested, in which: 6 groups of parallel tests were conducted once a day for 3 days to verify the repeatability and precision of the detection method of the present disclosure; the detection results of the three brands of commercial formula milk powder were shown in Tables 6-8, in which, LD represents the characteristic peptide segment of α -lactalbumin, LDQWLCEK (SEQ ID NO. 1), TP represents the characteristic peptide segment of β -lactoglobulin, TPEVDDEALEK (SEQ ID NO. 3), FAL represents the characteristic peptide segment of α -casein, FALPQYLK (SEQ ID NO. 4), and VL represents the characteristic peptide segment of β -casein, VLPVPQK (SEQ ID NO. 2).

(76) TABLE-US-00016 TABLE 6 Repeatability and precision results of the detection of 1# brand formula milk powder (Sanyuan, Enbeiyi, Section 1) Days 1 2 3 TP/FAL LD/VL TP/FAL LD/VL TP/FAL LD/VL 5.661328 0.596471 5.943357 0.593253 5.854161 0.594823 6.698231 0.591934 6.182401 0.636411 6.215530 0.601842 5.833083 0.626097 6.463053 0.658798 6.251452 0.620248 6.070490 0.592644 5.611366 0.614108 5.950025 0.581455 6.191200 0.604171 5.853564 0.582013 6.072519 0.562768 6.120052 0.558423 6.115780 0.597187 6.182007 0.561523 Average 6.095731 0.594957 6.028253 0.613628 6.087616 0.587110 SD 0.355257 0.021937 0.293936 0.029136 0.158603 0.023038 RSD (%) 0.058280 0.036871 0.04876 0.047482 0.026053 0.039241 Average 6.070533 0.598565 between groups SD 0.036840 0.013623 RSD (%) 0.006069 0.022759

(77) TABLE-US-00017 TABLE 7 Repeatability and precision results of the detection of 2# brand formula milk powder (Sanyuan, Aixinbao, Section 1) Days 1 2 3 TP/FAL LD/VL TP/FAL LD/VL TP/FAL LD/VL 6.255670 0.611835 5.629679 0.615912 5.666182 0.586044 5.760567 0.605571 5.619842 0.648726 5.724010 0.538855 6.331991 0.590981 5.304704 0.630166 5.634282 0.574862 6.156652 0.634878 5.836408 0.631432 5.471585 0.574179 5.802848 0.562343 5.592876 0.613239 5.762156 0.532703 5.901041 0.552298 5.636404 0.627743 5.465830 0.576933 Average 6.034795 0.592984 5.603319 0.627870 5.620674 0.563929 SD 0.244480 0.031200 0.170689 0.012728 0.125822 0.022300 RSD (%) 0.040512 0.052615 0.030462 0.020272 0.022386 0.039545 Average 5.834348 0.605599 between groups SD 0.244257 0.032015 RSD (%) 0.041865 0.052864

(78) TABLE-US-00018 TABLE 8 Repeatability and precision results of the detection of 3# brand formula milk powder (Sanyuan, Ailiyou, Section 1) Days 1 2 3 TP/FAL LD/VL TP/FAL LD/VL TP/FAL LD/VL 7.453970 0.531694 7.704645 0.561678 6.723771 0.593539 7.259978 0.595509 7.603946 0.570701 8.520871 0.662325 7.114793 0.510083 7.885622 0.638107 8.433776 0.676738 7.299271 0.476919 7.420642 0.548663 8.070415 0.759728 7.429340 0.553053 7.749015 0.577207 8.319099 0.720309 7.765910 0.583946 7.761472 0.602464 7.881502 0.613721 Average 7.387210 0.541867 7.687557 0.583137 7.991573 0.671060 SD 0.222611 0.044956 0.159372 0.032354 0.664624 0.062759 RSD (%) 0.030135 0.082965 0.020731 0.055483 0.083166 0.093522 Average 7.688780 0.598688 between groups SD 0.302183 0.065985 RSD (%) 0.039302 0.110217

(79) According to Tables 6-8, the RSD within groups for the above detections was in the range of 2.03%-9.35%, and the RSD between groups was in the range of 0.61%-11.02%, indicating that the detection method of the present disclosure had good repeatability and precision.

Example 6: Detection Example of the Detection Method of the Present Disclosure

(80) According to the detection method of the present disclosure described in Example 3, the content of whey protein in commercial formula milk powder (Abbott, Classic Enmeili, Section 1)

was detected, and the specific procedure was as follows:

(81) To 0.2 g commercial formula milk powder (Abbott, Classic Enmeili Section 1), ultrapure water was added to a fixed volume of 200 mL. The obtained sample solution was then filtered roughly with pure acetic acid filter membrane with a pore size of 0.45 μm . Then 250 μL of the filtered sample solution was added to 150 μL of ammonium bicarbonate solution, followed by adding 10 μL of dithiothreitol solution, and kept in a water bath at 70° C. for 30 min. After cooling to room temperature, 30 μL of iodoacetamide solution was added thereto, and allowed to stand for 30 min in the dark. After illuminating for 10 min, 10 μL of calcium chloride solution was added thereto, followed by adding 50 μL of Trypsin solution to allow an enzyme digestion at 37° C. for 28 h, and further followed by adding 10 μL of acetic acid solution and standing for 15 min to stop digestion. Filtration was carried out with 0.22 μm of polyethersulfone filter membrane prior to detection on the machine. It was calculated that whey protein accounted for 61.77% of the total protein content in the sample.

(82) Finally, it should be noted that the above embodiments are only used to illustrate the technical solutions of the present disclosure, but not to limit it; although the present disclosure has been described in detail with reference to the foregoing embodiments, it will be understood by one of ordinary skill in the art that the technical solutions described in the foregoing embodiments can still be modified or some technical features can be equivalently substituted; however, these modifications or substitutions do not make the essence of the corresponding technical solutions departing from the spirit and scope of the technical solutions of various embodiments of the present disclosure.

Claims

1. A method for detecting the contents of whey protein and casein, and/or the ratio thereof in milk powder by liquid chromatography-mass spectrometry, comprising: 1) Preparing a standard solution of whey protein with the characteristic peptide segments of α -lactalbumin and β -lactoglobulin, in accordance with the α -lactalbumin and β -lactoglobulin accounting for 20% and 50% of the whey protein, respectively; 2) Preparing a standard solution of casein with the characteristic peptide segments of α -casein and β -casein, in accordance with the α -casein and β -casein accounting for 50% and 40% of the casein, respectively; 3) Mixing the standard solution of whey protein from step 1) with the standard solution of casein from step 2) to prepare a series of mixed protein standard solutions with a series of whey protein:casein ratios; 4) Drawing standard curves with the ratio of whey protein to casein as abscissa and the peak area ratios of characteristic peptide segments of α -lactalbumin to β -casein and the peak area ratios of characteristic peptide segments of β -lactoglobulin to α -casein as ordinate; 5) Detecting the milk powder to be tested for the peak area ratios of the characteristic peptide segments of β -lactoglobulin to α -casein and the peak area ratios of the characteristic peptide segments of α -lactalbumin to β -casein; 6) Substituting the peak area ratio of the characteristic peptide segments of β -lactoglobulin to α -casein in the detected milk powder into the standard curve to determine the ratio M of whey protein to casein, and calculate the actual amounts of β -lactoglobulin and α -casein based on the ratio M thus determined; and, substituting the peak area ratio of the characteristic peptide segments of α -lactalbumin to β -casein in the detected milk powder into the standard curve to determine the ratio N of whey protein to casein, and calculate the actual amounts of α -lactalbumin to β -casein based on the ratio N thus determined; 7) Acquiring the actual contents of whey protein and casein, and/or the actual ratio of whey protein/casein based on the actual amounts of α -lactalbumin, β -casein, β -lactoglobulin, and α -casein obtained from step 6).

2. The method according to claim 1, wherein the characteristic peptide segments of α -lactalbumin, β -casein, β -lactoglobulin, and α -casein are LDQWLCEK shown in SEQ ID NO. 1, VLPVPQK shown in SEQ ID NO. 2, TPEVDDEALEK shown in SEQ ID NO. 3, and FALPQYLK shown in

SEQ ID NO. 4, respectively.

3. The method according to claim 1, wherein in step 3), the series of whey protein:casein ratios include whey protein:casein ratios of 0.25, 0.43, 0.67, 1, 1.5, and 2.33, respectively.

4. The method according to claim 1, wherein in step 6), the actual amounts of β -lactoglobulin and α -casein are calculated by the following formulas, respectively:

- lactoglobulin = $\frac{M1}{M1+M2}X * 50\%$, - casein = $\frac{M2}{M1+M2}X * 50\%$; wherein M1 is the preceding term in the ratio M of whey protein:casein as determined based on the peak area ratio of characteristic peptide segments of β -lactoglobulin to α -casein, and M2 is the latter term in the ratio M of whey protein:casein as determined based on the peak area ratio of characteristic peptide segments of β -lactoglobulin to α -casein, with the provisions of: $M=M1/M2$, and $M1+M2=1$; and wherein X is the total protein concentration of the sample.

5. The method according to claim 1, wherein in step 6), the actual amounts of α -lactalbumin and β -casein are calculated by the following formulas, respectively:

- la = $\frac{N1}{N1+N2}X * 20\%$, - cs = $\frac{N2}{N1+N2}X * 40\%$; wherein N1 is the preceding term in the ratio N of whey protein:casein as determined based on the peak area ratio of characteristic peptide segments of α -lactalbumin to β -casein, and N2 is the latter term in the ratio N of whey protein:casein as determined based on the peak area ratio of characteristic peptide segments of α -lactalbumin to β -casein, with the provisions of: $N=N1/N2$, and $N1+N2=1$; and, wherein, X is the total protein concentration of the sample.

6. The method according to claim 1, wherein, in step 7), the formula for calculating the actual content of whey protein is: $\frac{\frac{N1}{N1+N2}X * 20\% + \frac{M1}{M1+M2}X * 50\%}{0.7}$ the formula for calculating the actual content of casein is: $\frac{\frac{M2}{M1+M2}X * 50\% + \frac{N2}{N1+N2}X * 40\%}{0.9}$ the formula for calculating the actual ratio of whey protein/casein is:

$\frac{\frac{N1}{N1+N2}X * 20\% + \frac{M1}{M1+M2}X * 50\%}{0.7} / \frac{\frac{M2}{M1+M2}X * 50\% + \frac{N2}{N1+N2}X * 40\%}{0.9} = \frac{18N1 + 45M1}{35M2 + 28N2} = \frac{63MN + 18N + 45M}{35N + 28M + 63}$; wherein M1 is the preceding term in the ratio M of whey protein:casein as determined based on the peak area ratio of characteristic peptide segments of β -lactoglobulin to α -casein, and M2 is the latter term in the ratio M of whey protein:casein as determined based on the peak area ratio of characteristic peptide segments of β -lactoglobulin to α -casein, with the provisions of: $M=M1/M2$, and $M1+M2=1$; wherein N1 is the preceding term in the ratio N of whey protein:casein as determined based on the peak area ratio of characteristic peptide segments of α -lactalbumin to β -casein, and N2 is the latter term in the ratio N of whey protein:casein as determined based on the peak area ratio of characteristic peptide segments of α -lactalbumin to β -casein, with the provisions of: $N=N1/N2$, and $N1+N2=1$; and wherein X is the total protein concentration of the sample.

7. The method according to claim 1, wherein, the detection in step 5) comprises the steps of sample treatment and enzyme digestion: dissolving the milk powder sample to be tested into a sample solution with a protein concentration of 0.1-0.4 mg/ml; subjecting the sample solution to rough filtration with a pure acetic acid filter membrane with a pore size of 0.45 μ m; adding the roughly filtered sample solution into ammonium bicarbonate solution, followed by adding dithiothreitol solution, and standing in a water bath at 65-75° C. for 25-35 min; after cooling, adding iodoacetamide solution, and standing for 25-35 min in the dark; illuminating the mixture for 8-12 min, followed by adding calcium chloride solution; then adding trypsin solution to allow an enzyme digestion at 37° C. for 26-30 h; then adding acetic acid solution to stop the enzyme digestion; filtering the reaction mixture with polyethersulfone filter membrane, and then performing selective ion scanning analysis by mass spectrometry.

8. The method according to claim 1, wherein the upper detection limit of the method is at a total protein concentration of 0.4 mg/mL.

9. A combination of characteristic peptide segments for detecting the ratio of whey protein:casein by liquid chromatography-mass spectrometry, comprising: characteristic peptide segments of α -lactalbumin, β -casein, β -lactoglobulin, and α -casein with amino acid sequences as shown in SEQ

ID NO. 1, SEQ ID NO. 2, SEQ ID NO. 3, and SEQ ID NO. 4, respectively.

10. The combination of characteristic peptide segments according to claim 9, which consists of the characteristic peptide segments of α -lactalbumin, β -casein, β -lactoglobulin, and α -casein with amino acid sequences as shown in SEQ ID NO. 1, SEQ ID NO. 2, SEQ ID NO. 3, and SEQ ID NO. 4, respectively.
